DOMINICK BRAICO

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Education

University of Michigan Ann Arbor Dec 2027
M.S. Robotics

University of Illinois Urbana-Champaign May 2026
B.S. Mechanical Engineering, Minor in Computer Science GPA: 3.91

Technical Skills

Simulation & Robotics: ROS 2, Isaac Sim, Gazebo, MuJoCo, PX4
Languages/Libraries: C/C++, Python (PyTorch, NumPy, JAX), MATLAB/Simulink
Developer Tools: Git, Linux, Docker, VS Code, Claude Code

Experience

Graduate Robotics Research Assistant Aug 2026 – Present
Michigan Robotics - Distributed Autonomous Systems and Control Lab
- Architecting a network-physics co-simulator coupling ns-3 with multi-agent robot simulation to model inter-robot wireless links with configurable latency, packet loss, and dynamic link drops
- Developing ROS 2 bridges between PX4 SITL and Isaac Sim, enabling a single offboard control stack to run unchanged across simulators
- Implementing decentralized multi-robot control algorithms as C++/Python ROS 2 packages, validated in simulation under realistic wireless communications constraints before deployment on hardware

Robotics Research Intern May 2026 – Aug 2026
Carnegie Mellon University Robotics Institute - Matt Lab
- Owned hardware architecture and a $90k procurement budget for an autonomous server-disassembly workcell, selecting and integrating collaborative arms, force/torque sensors, RGB-D cameras, and a servo-driven conveyor
- Designed the workcell's full mechanical system, including custom compliant end-effectors swapped via tool changer to remove 6 distinct blade-server components

Controls Engineering Intern May 2025 – Aug 2025
Whisper Aero
- Developed FOC firmware in C for 3-phase BLDC controller boards, with CAN bus communication to a custom Python GUI for motor control and auto-tuning
- Automated eddy-current dynamometer testing of custom BLDC motors in Python, characterizing performance for control validation and significantly cutting manual motor test time

Robotics R&D Intern May 2024 – Mar 2025
Construction Engineering Research Laboratory - Robotics for Engineer Operations
- Built the electrical and compute enclosure that enabled single-operator control of Co3MaNDR, a novel cable-driven parallel robot, integrating onboard compute, motor drivers, and power distribution for its cable actuators

Projects

Quadruped Reinforcement Learning
- Trained a velocity-conditioned walking policy for a 12-DoF Barkour quadruped using PPO in/MuJoCo MJX on GPU
- Implemented body-frame linear and yaw velocity tracking rewards within an 11-term reward function

Robotic Mobile Manipulator Simulation
- Engineered a custom mobile manipulator physics model in Gazebo featuring a Clearpath Husky mobile platform integrated with a UR3 manipulator arm and Robotiq 2F-85 gripper

Autonomous Surface Vessel
- Implemented and tuned a PID heading controller for a vessel with rudder and differential-thrust steering, holding heading within ±2° at steady state

GHOST Electric Motorcycles
- Led motor control and sensor integration for a 103.6 V PMAC powertrain on the team's first motorcycle, designing and testing control systems prior to vehicle integration
- Configured CAN bus network for communication between motor controller, BMS, and onboard sensors; performed bench testing to validate electrical system performance prior to vehicle integration